

Problem Solution Fit

Project Title: Rising Waters

Flood Prediction System Using Machine Learning

Introduction

Floods cause serious damage to people, property, agriculture, and transportation. Many flood warnings are issued late because traditional methods rely mainly on manual observation and historical data. An intelligent prediction system can help provide faster and more reliable flood alerts.

Problem

Existing flood prediction methods are often time-consuming and may not provide early warnings. People and authorities require a system that can quickly analyze weather conditions and predict flood risk before the situation becomes dangerous.

Proposed Solution

The Rising Waters system uses Machine Learning to analyze weather-related parameters such as temperature, humidity, cloud cover, and rainfall values. The trained model predicts whether flood conditions are likely and displays the result through a simple web application.

Problem–Solution Fit

The proposed solution directly addresses the existing problem by replacing manual analysis with an automated prediction system. It provides quick predictions, improves decision-making, and helps users receive early flood warnings.

Benefits

- Provides early flood prediction
- Reduces disaster risk
- Improves prediction accuracy
- Supports disaster management
- Easy to use through a web application

Conclusion

The Rising Waters system effectively matches the identified problem with an intelligent Machine Learning solution. It offers faster, more accurate, and user-friendly flood prediction to improve public safety.